Risk-Adjusted Determinants of Prolonged Hospital Stay for surgical procedures regarding cerebrovascular care

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1 Introduction

Cerebrovascular disease, commonly known as stroke or cerebrovascular accident, is a medical emergency characterized by an acute interruption of blood flow to the brain. Two main types can be distinguished: ischemic stroke, which accounts for about 85% of cases and is caused by an obstruction of blood flow, and hemorrhagic stroke, caused by the rupture of a blood vessel. Among the risk factors for ischemic stroke is a high level of low-density lipoproteins (LDL), which promotes the formation of atherosclerotic plaques in the arteries, including the carotid arteries; plaque buildup can reduce blood flow or form clots, increasing the risk of thrombotic or embolic stroke. Acute cerebrovascular diseases are a frequent cause of hospital admission and require rapid and appropriate interventions to reduce the risk of serious complications; among these, carotid endarterectomy and stenting are commonly performed surgical procedures to treat carotid stenosis and prevent ischemic strokes, especially in patients who have experienced symptoms such as a transient ischemic attack (TIA).

The two procedures differ in their approach: carotid endarterectomy is a surgical operation that physically removes plaque from the artery wall, while stenting is a less invasive procedure that involves placing a metal mesh to keep the artery open. There have been various studies on the performance of the two interventions and which is the best is an argument still in debate, as a consequence the two are under the same ICD-10 procedure code, so they will be analyzed together in our study, since it is not the objective of this study to back up one of the two procedures.

In recent years, the focus of studies following certain medical procedures has expanded beyond clinical effectiveness to include the patient experience as an independent dimension of care quality, moreover when the procedures under study are made as prevention to further complication or reduce the risk of more dangerous diseases. Focusing on the patient's perceptions and needs helps improve treatment effectiveness, guide research, and inform health policies through a more personalized approach. Length of hospital stay (LOS) can serve as an objective indicator of care quality. Traditionally seen as a measure of efficiency and cost, LOS can also provide information on patient safety and the adequacy of care: a prolonged stay may signal quality problems, as poor care tends to generate complications that extend hospitalization.

This study aims to investigate how hospitals in New York State differ in the management of patients with acute cerebrovascular disease undergoing surgical procedures, focusing in particular on the length of hospital stay. Specifically, it will examine how the proportion of patients whose stay exceeds the 75th percentile varies across hospitals, taking into account patient characteristics, disease severity, and type of admission. Another goal is to identify which hospitals show excess length of stay after surgical interventions, comparing results after standardizing crude rates of prolonged LOS. Finally, the study aims to assess the impact of carotid endarterectomy or stenting on patient outcomes, compared with those who did not undergo these procedures, accounting for baseline patient characteristics using propensity score methods.

2 Materials

The dataset that has been examined comes from a merge of New York dataset of hospital inpatient discharges provided by the Statewide Planning and Research Cooperative System (SPARCS) from 2022 to 2024. Each entry is a patient that has been discharged by a hospital in the New York area, for whom patient-level information such as demographic characteristics, diagnoses, procedures, length of stay, discharge status, payer type, and hospital identifiers are provided. Diagnoses and procedures are coded

using ICD-10 classifications. To carry out the analysis our population is comprised of people who had been diagnosed with a diagnosis linked to cerebrovascular diseases (i.e. acute hemorrhagic cerebrovascular disease, cerebral infarction, occlusion or stenosis of precerebral or cerebral arteries without infarction, other and ill-defined cerebrovascular disease, sequela of cerebral infarction and other cerebrovascular disease, sequela of hemorrhagic cerebrovascular disease and transient cerebral ischemia) and was treated with a surgical procedure. Among the ones who received surgery, people who died in the following days while they were in hospital care have been omitted from the study, in order to account for the downward bias on the length of stay. After all the total number of patients considered was equal to 54992 people, among these, 8820 have undergone endarterectomy or stenting.

3 Methodologies

First we explored the demographic characteristics and made descriptive analysis using frequency tables. The length of stay is the variable of interest to study performance of the hospitals; the original count variable was transformed into a binary indicator, defined as 1 for observations with values above the third quartile, that was equivalent to 11 days, and 0 otherwise. Wilcoxon rank sum test and odds ratio were used to study whether a longer stay in hospital was significantly correlated with the surgical procedures under study.

To account for the clustering of patients within hospitals, we fitted a GEE model with a logit link function, acknowledging that patients treated in the same hospital are not independent and measures are likely to be correlated. The full model includes all relevant covariates focused on patient and admission characteristics, to evaluate their association with prolonged length of stay (LOS).The correlation structure that has been used was the exchangeable one, where an estimate of the average of all correlation among clusters was considered. However, with the GEE model we cannot see how the length of stay varies among hospitals. Multilevel models can effectively incorporate different sources of variations, including measures of heterogeneity that can support application of this approach. A multilevel logit model with random intercept was used to quantify how much of the variation in prolonged length of stay is attributable to differences between hospitals, using measures such as Intraclass correlation coefficient (ICC) and Median Odds Ratio (MOR). While dealing with modelization we excluded all hospitals with a volume of cases lower than 100, to avoid inflated results with specific hospital variation.

$$ICC = \frac{\tau^2}{\tau^2 + \sigma^2} \qquad\qquad MOR = \exp(\sqrt{2\tau^2 x \phi_{0.75}^{-1}})$$

Backward elimination approach was performed on both full models to obtain a more parsimonious model. From the set of all covariates that we deemed to be linked with length of hospital stay, we eliminated one at time the variables that were not significant. Then we compared for both type of modelling the reduced and full model to choose the better one, for the mixed-effect models we used the likelihood-ratio test, while for GEE models we compared the QIC statistic . A model selection criterion that replaces the AIC while dealing with GEE models, due to the fact that only quasi-likelihood is available.

Then we performed direct and indirect standardization to compare which hospitals showed a longer stay in hospital after surgical intervention accounting for the different population composition among the hospitals with highest volume. In order to perform direct standardization the overall dataset was considered as a standard population and then we calculated age-sex specific rates of patients who were in care longer than 11 days with respect to those who were not and weighted by the structure of the standard population.

$$DSR = \frac{\sum_{i,j} N_{ij} \frac{d_{ij}}{n_{ij}}}{N}$$

For the indirect standardization, the risk-adjusted rate was calculated as the ratio of observed patients who had a long stay over the expected number if they had the same age-sex rate of the standard population according to a fitted logit function with age and sex as predictors, multiplied by the reference population rate.

$$RAR_i = \alpha \frac{\frac{\sum_{ij} Y_{ij}}{n}}{\frac{\sum_{ij} P_{ij}}{n}} = \alpha \frac{O_i}{E_i}$$

Lastly, propensity scores were applied to the dataset to adjust for systematic differences between patients undergoing endarterectomy or stenting and those undergoing other surgical procedures. The propensity scores were estimated as the predicted probability of receiving endarterectomy or stenting, using a logistic regression based on the patient's demographic, clinical, and procedural characteristics. The propensity scores were then used as weights in a GEE (Generalized Estimating Equations) model following the Inverse Probability of Treatment Weighting (IPTW) method, in order to balance the baseline characteristics between treatment groups and to estimate the Average Treatment Effect (ATE) at the population level. This approach also allows all observations to be retained, preserving the sample size and the statistical power of the study. Some patients had propensity scores very close to 0, indicating that, based on their baseline characteristics, they had a very low probability of receiving endarterectomy or stenting. In the IPTW framework, such patients receive higher weights to ensure proper balance between treatment groups. To limit the influence of extreme weights and improve the stability of the estimates, weights[1] were truncated at the 1st and 99th percentiles.

$$e_i = Pr(Z_i = 1|X_i)$$

$$P = \frac{e^{a+bX}}{1+e^{a+bX}} \qquad\qquad w_i = \frac{Z_i}{e_i} + \frac{1-Zi}{1-e_i}$$

4 Results

If we look overall at the funnel plot (figure 1), there are many hospitals which are over the confidence interval towards the left-hand side of the plot, but this is expected since they have a small sample and thus have a higher variability. What is interesting to note is that, looking at higher volume facilities, there might a trend linked to the age structure of the patients that are in care. If we look at the red dots that are hospitals with a population comprised of less than a fifth of elderly people, to have better performance regarding length of stay since all place lower than the mean of crude rate.

As shown in table 1, the risk-adjusted analysis of the six hospitals with patient populations exceeding 1,000 reveals substantial variability in performance among them. After obtaining age-sex specific standardized rates it is possible to observe two important facts: first if we compare the top 6 hospitals, hospital 87 performs better with respect to all the other stand_rates, second that hospital 6 has a great discrepancy between the crude rate and the standardized one, that means if we apply the age-sex composition of the standard population the hospital would perform much worse, the rate skyrockets from a mere 9% to 29.52% of the population that would have a longer stay in hospital. In fact if we look at the demographic composition of patients from hospital 6, there is a higher number of people younger than 50 years old with respect to the other facilities.

Even when we look at risk-adjusted rate by age-sex composition, we can see that hospital 87 perform greatly, the RAR is only 5.91% with respect to the expected rate of 16.29%; meanwhile hospital 182 had a more complex case mix, represented by the expected rate of 29.02, but performed even worse with a RAR of 34.78. But, there is a notable discrepancy between standardized results and Risk-adjusted rates for Auburn Community Hospital, since their RAR is lower than the expected number of cases of overstaying, it means that the hospital performs well in handling their current patient load, on the other hand the hospital's stratum-specific results reveal weaker performance in certain patient subgroups. As noted previously, outcomes among elderly patients appear to be a key factor contributing to the lower standardized rate. This subgroup disproportionately affects the overall standardized results, thereby masking the hospital's otherwise strong performance reflected in the risk-adjusted metrics. It is important to note, however, that this apparent under-performance among elderly patients may be influenced by the relatively small sample size within this specific cluster. As a result, the discrepancy between the standardized and risk-adjusted measures should be interpreted with caution, particularly when evaluating performance within smaller patient subgroups.

Regarding the full GEE model, the variable representing gender ("males") is not significant with a p-value of 0.10373, suggesting limited contribution to explaining variation in the length of stay. Comparing the full and reduced model through QIC we found out that the two values were similar, so a reduced model can be selected since it is more parsimonious. It is apparent that the strongest association with

[1]Weight truncation (usually at the 1st and 99th percentiles) is applied to limit the influence of extreme weights and reduce the variance of the estimate; however, this procedure can introduce bias because it modifies the reference population. Chesnaye, N. C., et al. (2021)

prolonged LOS is severe conditions, followed by risk. Patients classified as "severe" have 3.13 times higher odds of experiencing a prolonged length of stay and patients classified as "risky", have 1.9 times higher odds (table 2). To understand better hospital differences, we fitted a multilevel logistic regression model. As in the GEE model, the variable "males" was not significant, and "severe" and "risk" presented the highest association with the length of stay. The likelihood ratio test as the case before the criterion favored the reduced nested model.

The null multilevel logistic regression model (who included only the intercept) and the model consisting of patient characteristics, showed that there is significant variation between hospitals in prolonged length of stay. At an average hospital, the probability of a patient who underwent surgery having a prolonged length of stay was 0.5696. Estimating random effects, the null model had an estimation of the between-hospital variance of 0.6172 and 0.2342 for the other model. These correspond to ICC of 0.158 and 0.0664, respectively. For the null model, 15.8% of the individual variation in the underlying propensity to stay longer in hospital care is due to systematic differences between hospitals, while the remaining 84.2% is attributable to patient-level factors and unobserved variation. According to the model consisting of patient characteristics, 6.63% is due to differences between hospitals, suggesting that a substantial portion of the initial hospital-level variation was driven by patient case-mix. For the model consisting of patient characteristics, if we look at the MOR, two identical patients at randomly selected hospitals would have a 1.59 difference in their odds of prolonged LOS solely due to the hospital they attended (down from 2.12 in the null model). Overall, the multilevel logistic model confirms that accounting for hospital-level clustering is essential. Patients within the same hospital share factors that influence the prolonged length of stay, and ignoring this clustering could underestimate standard errors.
Then we plotted the distribution of propensity scores (figure 2) in the two groups (patients treated with endarterectomy or stenting, endar = 1, and patients undergoing other surgical procedures, endar = 0). The propensity score represents the estimated probability that a patient receives the treatment (endarterectomy or stenting) based on observed baseline characteristics. A strong concentration of low propensity score values (ps < 0.2) is observed, particularly in the untreated group (endar = 0), indicating that a substantial proportion of patients had a very low probability of receiving endarterectomy. The treated group (endar = 1) shows a more dispersed distribution, with a greater presence of patients with medium-high propensity scores, consistent with a higher estimated probability of treatment. Nevertheless, a substantial area of overlap between the two groups is present, particularly between 0.0 and 0.7, suggesting the existence of a comparable population of patients for whom treatment assignment could have plausibly occurred. From the value of propensity scores we computed the weights as the inverse of the PS and applied them on the GEE model.

The IPTW-weighted GEE model shows that patients who underwent endarterectomy or stenting have a significantly lower probability of experiencing a long hospital stay compared with patients who didn't receive the treatment. The estimated coefficient is -1.07, corresponding to an odds ratio of approximately 0.34, which indicates about a 66% reduction in the probability of having a long hospital stay. All variables included in the model are statistically significant. Overall, these results suggest that, after balancing baseline characteristics using IPTW weights, treatment with endarterectomy or stenting is associated with a significant reduction in the risk of prolonged hospital stay, confirming the beneficial impact of the treatment on hospital outcomes.

5 Conclusion

In conclusion, we found out that the rate of prolonged stay is mainly driven by patient-level clinical factors; nevertheless, significant institution variability is persistent across the healthcare system. GEE showed that, across all hospitals, clinical severity and mortality risk were the most significant contributors in the proportion of patient with prolonged stay. Having a MOR=1.59 in the multilevel model means that a patient's probability of overstaying is significantly influenced by the specific hospital they attend, independent of their health status and even by exploring the single hospital variability through mixed effect models these factors were both deemed significant. Thus while the assessment of patient conditions seems to be the key in a good management of patients with cerebrovascular diseases, there is room for improvement inside each single hospitals. As we have seen there are many differences in outcome even within the largest hospitals.

Regarding this subject, risk adjustment was a key factor in evaluating the hospital performances, since the results of prolonged stay rate obtained by the Auburn community hospital would have been easily misinterpreted. Being by far the largest hospital by volume among the population, extreme care

must be taken when we are looking at its numbers. By just looking at the crude rate, it would easily interpreted as one of the best performance facilities, but after standardizing the results the situation was much more complex. Direct and indirect standardized rate tell us that while the hospital performed good with the pool of cases they treated, it was also because they had a favorable patient age-sex distribution and by further looking into it we found that the number of elderly people was drastically lower than the others. So low that the per strata correction would not be significant.
In total we looked at the effect of endarterectomy and stenting as a whole, as expected, we found that the two lead to a significant decrease in the proportion of patients who had a long stay in hospitals. The application of Inverse Probability of Treatment Weighting on GEE model showed a 66% reduction in the odds of prolonged stay, this means that both procedures, despite the long-lasting debate on the better one, are resource effective procedures for hospitals. As time passes both interventions are becoming more and more common in the prevention of ischemic stroke and, while focusing on the main issue that is the risk of death or invalidity due to sudden cerebral diseases, hospitals are needed to take into account the well-being of their patient and to adopt the most efficient measures to ensure a better treatment care for all. So studying performance indicators such as length of hospital stay even for other kind of procedures and diagnosis would give great benefits for both patients and medical establishments.

Future studies should include a broader and more specific set of covariates, we were not able to implement any kind of clinical characteristics of the patient, as the dataset included in the study was open-source it was not possible to study the procedures in a medically accurate way, that would have led to a better estimate of the propensity scores. Another limitation of the study was that we were not able to identify causes of hospital-level variation, a Difference in Differences approach based on hospital policies would be ideal in this kind of study, but, as before, the limitation of data had not allowed us to further explore the issue. Furthermore the propensity scores showed a skewed distribution to the left, a set of more clinical related variable could better explain the treatment allocation.

6 Tables and Figures

Table 1: Hospital Rate Analysis Data

Unit	**Name**	**Crude Rate**	**Stand. Rate**	**Stand. CI**	**N**	**Exp. Rate**	**Risk Adj. Rate**	**Risk Adj. CI**
Hospital 6	Auburn Community	9.09	29.52	(29.5 – 29.53)	2232	14.91	13.43	(11.18 – 15.67)
Hospital 94	Montefiore-Wakefield	15.69	27.02	(27 – 27.04)	1428	18.65	20.17	(17.6 – 22.75)
Hospital 102	Mt Sinai - Queens	7.39	20.95	(20.93 – 20.98)	1340	17.14	9.89	(7.16 – 12.61)
Hospital 87	Memorial Cancer	4.22	20.85	(20.82 – 20.87)	1113	16.29	5.91	(2.83 – 8.99)
Hospital 182	Stony Brook Univ.	39.91	41.72	(41.69 – 41.74)	1060	29.02	34.78	(32.46 – 37.1)
Hospital 129	NYU Langone-LI	22.47	23.71	(23.69 – 23.74)	1037	23.11	23.71	(21.07 – 26.36)

Table 2: Association Estimates from GEE and GLMER Models

Variable	**GEE**			**GLMER**		
	OR	**95% CI**	**p-value**	**OR**	**95% CI**	**p-value**
recl_age	0.937	(0.904 – 0.971)	< 0.001	0.934	(0.904 – 0.971)	< 0.001
race	0.813	(0.792 – 0.835)	< 0.001	0.831	(0.792 – 0.835)	< 0.001
year	0.542	(0.513 – 0.572)	< 0.001	0.511	(0.513 – 0.572)	< 0.001
adm_type	1.261	(1.226 – 1.297)	< 0.001	1.272	(1.226 – 1.297)	< 0.001
procedure	1.003	(1.003 – 1.004)	< 0.001	1.003	(1.003 – 1.004)	< 0.001
risky	1.898	(1.801 – 2.001)	< 0.001	1.958	(1.801 – 2.001)	< 0.001
severe	3.127	(2.943 – 3.321)	< 0.001	3.232	(2.943 – 3.321)	< 0.001

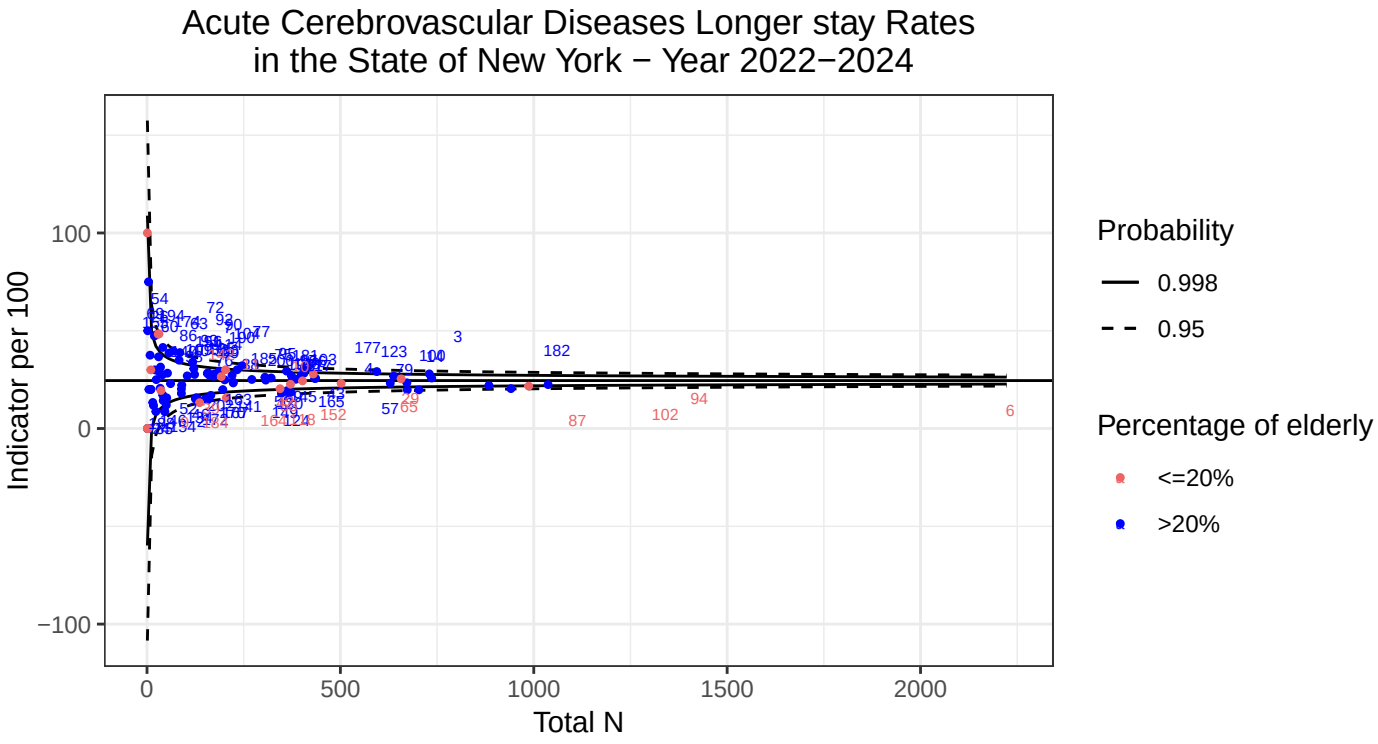


Figure 1: Funnel Plot: prolonged stay of patients with surgical procedure in New York, 2022-2024

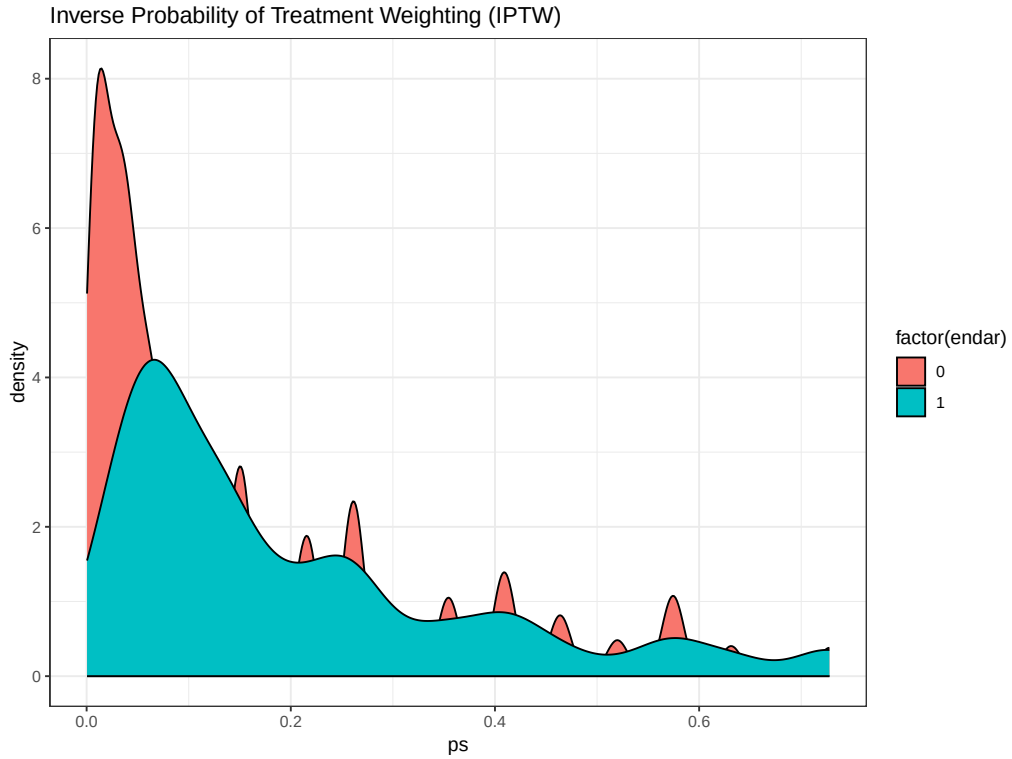


Figure 2: Distribution of propensity scores

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